Investigation: Grupo Casas Bahia S.A. (CVM code 6505), fiscal year 2023, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Comércio (Atacado e Varejo) (origin: FCA); group: non-financial.
Trading codes (2023 FCA, version 2, filed on 2023-10-02): BHIA12 (Bônus de Subscrição, Bolsa, since 2023-09-18); BHIA3 (Ações Ordinárias, Bolsa, since 2018-11-26).

Document used

2023 DFP, version 2, filed on 2024-03-25; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2024-03-25); 2 (2024-03-25).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=135112

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −3.823 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 8 (n = 42)
- Percentile in group and year: 7 (n = 530)
- The company's own history: median −2.680; Tukey range (1.5 × IQR): from −3.872 to −0.800; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 0.583 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 6 (n = 44)
- Percentile in group and year: 8 (n = 599)
- The company's own history: median 0.956; Tukey range (1.5 × IQR): from 0.310 to 2.011; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.068 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 88 (n = 44)
- Percentile in group and year: 73 (n = 609)
- The company's own history: median 1.041; Tukey range (1.5 × IQR): from 0.810 to 1.243; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.169 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 60 (n = 44)
- Percentile in group and year: 84 (n = 610)
- The company's own history: median 1.120; Tukey range (1.5 × IQR): from 0.725 to 1.428; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

SGI, sales growth index
Value: 0.934 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 19 (n = 44)
Percentile in group and year: 21 (n = 610)
The company's own history: median 1.050; Tukey range (1.5 × IQR): from 0.896 to 1.173; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DEPI, depreciation index
Value: 0.896 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 30 (n = 42)
Percentile in group and year: 33 (n = 549)
The company's own history: median 0.930; Tukey range (1.5 × IQR): from 0.746 to 1.124; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 1.066 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 69 (n = 44)
Percentile in group and year: 61 (n = 606)
The company's own history: median 0.985; Tukey range (1.5 × IQR): from 0.666 to 1.362; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.208 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 8 (n = 44)
Percentile in group and year: 8 (n = 628)
The company's own history: median −0.040; Tukey range (1.5 × IQR): from −0.178 to 0.157; value below the range (index; 5 fiscal years)
Confidence in the data: medium

LVGI, leverage index
Value: 1.026 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 56 (n = 44)
Percentile in group and year: 61 (n = 619)
The company's own history: median 1.016; Tukey range (1.5 × IQR): from 0.946 to 1.066; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DPO jump
Value: −6.7 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 25.7 days)
Percentile in sector and year: 31 (n = 44)
Percentile in group and year: 29 (n = 591)
The company's own history: median 1.4 days; Tukey range (1.5 × IQR): from −44.8 days to 33.8 days; value inside the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix

(decision of 2026-11-07)

Implied interest rate minus average CDI
Value: −13.21 p.p.
Alert by the study 1 rule: no
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 2 (n = 40)
Percentile in group and year: 2 (n = 588)
The company's own history: median −5.94 p.p.; Tukey range (1.5 × IQR): from −9.49 p.p. to −1.45 p.p.; value below the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.197 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 10 (n = 44)
Percentile in group and year: 9 (n = 627)
The company's own history: median −0.041; Tukey range (1.5 × IQR): from −0.182 to 0.159; value below the range (index; 5 fiscal years)
Confidence in the data: medium

Financial risk (context only)

Altman Z" (financial risk, context only)
Value: −1.573 (index)
Components: X1 = −0.171; X2 = −0.102; X3 = −0.036; X4 = 0.121 (indices)
Percentile in sector and year: 10 (n = 44)
The company's own history: median −0.077; Tukey range (1.5 × IQR): from −0.452 to 0.379; value below the range (index; 5 fiscal years)
- No cut-off zone: the available source does not define zones for Z"
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 3/9 (9 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 1
- Change in ROA (ΔROA): 0
- Cash above earnings (ACCRUAL): 1
- Change in leverage (ΔLEVER): 1
- Change in current ratio (ΔLIQUID): 0
- No equity issuance (EQ_OFFER): 0
- Change in gross margin (ΔMARGIN): 0
- Change in asset turnover (ΔTURN): 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

What was known on the date

- DFP 2018, v1: 2019-02-20
- DFP 2019, v1: 2020-03-26
- DFP 2020, v1: 2021-03-02
- DFP 2021, v1: 2022-03-09
- DFP 2022, v1: 2023-03-09
- FCA 2023, v1: 2023-05-31
- FCA 2023, v2: 2023-10-02
- DFP 2023, v1: 2024-03-25
- DFP 2023, v2: 2024-03-25

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 0 failed.

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).
no opinion excerpt was found in the text
Auditor in the 2021 FCA: Ernst & Young Auditores Independentes S.S..
Auditor in the 2022 FCA: Ernst & Young Auditores Independentes S.S..
Auditor in the 2023 FCA: Ernst & Young Auditores Independentes S.S..
Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2014: Ernst & Young Auditores Independentes S.S. -> Deloitte Touche Tohmatsu Auditores Independentes. The outgoing firm audited 4 consecutive fiscal years (2010 to 2013), by the start and end dates in the FCA.
- 2018: Deloitte Touche Tohmatsu Auditores Independentes -> Ernst & Young Auditores Independentes S.S.. The outgoing firm audited 5 consecutive fiscal years (2014 to 2018), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).

- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 39 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E). Generated at 2026-11-10T19:41:34+00:00.

Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.